Lake Water Quality: Are We Winning the War?

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With this being the 50-year commemorative issue of The Michigan Riparian, it seems like a good time to ask the question, "Are we winning the war to preserve the quality of our lakes?" While this is a reasonable question, it is difficult to answer. Michigan has over 10,000 lakes greater than five acres, and collecting data from all of these waterbodies would be a formidable task. However, there are a number of sampling initiatives being coordinated at the federal, state, and local levels that make a general assessment of the condition of Michigan's lakes possible.

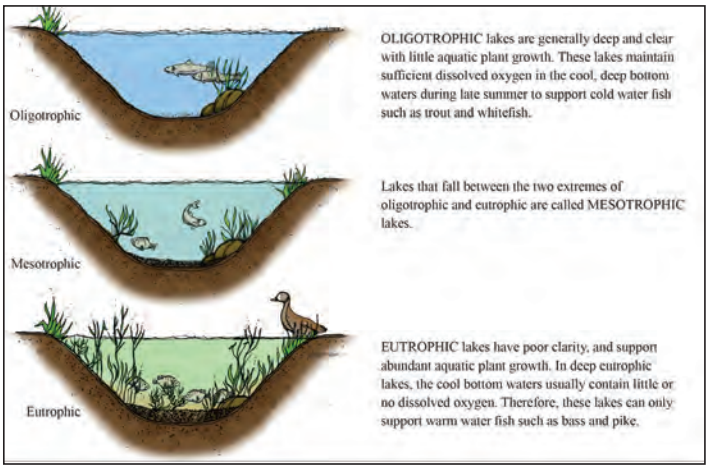
Lake Trophic Classification

One goal in assessing the condition of lakes is to classify lakes by "trophic state." Trophic state is a classification system based on a lake's ability to support plant and animal life. When classifying lakes, scientists use the broad categories "**oligotrophic**," "**mesotrophic**," or "**eutrophic**."

Under natural conditions, most lakes will ultimately evolve to a eutrophic state as they gradually fill with sediment and organic matter transported to the lake from the surrounding watershed. As the lake becomes shallower, the process accelerates. An increase in aquatic plant abundance also contributes to the lake filling in as sediment and decaying plant matter accumulate on the lake bottom. Eventually, terrestrial plants become established and the lake is transformed to a marshland. The natural lake aging process takes many thousands of years, but can be greatly accelerated if excessive amounts of sediment and nutrients (which stimulate aquatic plant growth) enter the lake from the surrounding watershed. Because these added inputs are usually associated with human activity, this accelerated lake aging process is sometimes referred to as "cultural eutrophication."

Key parameters used to classify lakes and to evaluate water quality include total phosphorus, chlorophyll-a, and Secchi transparency.

Phosphorus is the nutrient that most often stimulates aquatic plant growth and the rate at which a lake ages and becomes more eutrophic. By measuring phosphorus levels, it is possible to gauge the overall health of a lake.



Chlorophyll-a is a pigment that imparts the green color to plants and algae. A rough estimate of the quantity of algae present in the water column can be made by measuring the amount of chlorophyll-a in the water column.

A **Secchi disk** is a round, black and white, 8-inch disk that is used to estimate water clarity. Generally, plants can grow to a depth of about twice the Secchi disk transparency.

Trophic Characteristics of Michigan Lakes

A recent scientific report published by the United States Geological Survey (USGS) entitled Water Quality Characteristics of Michigan's Inland Lakes, 2001-10 (Fuller and Taricska 2012), provides an excellent overview of historical and recent sampling programs in Michigan. In this report, trophic classifications of 445 lakes sampled by the Michigan Department of Natural Resources and Environment between 1974 and 1984 were compared with a more recent data set derived from 729 lakes sampled by MDEQ and USGS between 2001 and 2010.

By examining phosphorus, chlorophyll-a, and Secchi disk transparency data, USGS determined that historic and recent lake trophic classifications were similar, with about 82 percent of lakes in the recent data set classified as mesotrophic or oligotrophic compared to about 79 percent of the lakes in the historic data set. Roughly speaking, about half of the lakes did not change trophic state; about one-quarter were more oligotrophic; and about one-quarter were more eutrophic. Thus, the bulk of Michigan lakes have good (mesotrophic) to excellent (oligotrophic) water quality, and water quality is largely unchanged today compared to historic data.

The Value of Water Quality Data

Michigan's water quality data presented in the USGS 2012 report are encouraging. There does not appear to be an overall shift toward degraded water quality statewide when comparing historical and more recent water quality data. Apparently, we are "holding our own." However, while some lakes showed improvement, others showed a shift to a more eutrophic state, which suggests that excessive nutrient loading is a problem on some lakes. Further, in many Michigan lakes, there is not sufficient data to discern long-term water quality trends on a "lake-specific" basis.

It is important to recognize that changes in lake water quality are often subtle and difficult to detect. There can be significant natural variability in lake water quality daily, seasonally and year-to-year. Because of this natural variability, it can be difficult to detect subtle changes or trends in water quality over time. In fact, it may take many years of regular sampling in a given lake to detect a statistically significant trend in water quality. This is why programs such as the Cooperative Lakes Monitoring Program are valuable (see Sampling Programs on page 25). By sampling an individual lake in a systematic manner over an extended period of time, it may be possible to detect subtle changes in water quality.

(Continued on page 25)

(Continued from page 23)

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Another consideration in attempting to measure changes in water quality is the concept of "response time." Lakes often do not respond immediately to a reduction in nutrient inputs. For lakes of average size and average water replenishment time, recovery may require 2-10 years (Wetzel 2001). If, for example, a sewer system is constructed around a lake, the improvement in water quality may not be immediately apparent and recovery may require several years. Again, this highlights the importance of collecting data over an extended time-period in order to document changes in water quality.

Recognizing the role of phosphorus in accelerating the lake eutrophication process, Michigan has placed limits on phosphorus in laundry and dishwasher detergents and lawn fertilizer. Over time, these limits may help to slow cultural eutrophication. The various monitoring programs and protocols that are now being implemented at the state and federal level are helpful in evaluating general water quality characteristics across Michigan. Hopefully, these types of programs will continue and, as additional data become available, we will be better able to evaluate long-term trends in water quality. However, funding for lake monitoring at the federal and state level is limited, and while overall statewide assessments of water quality may be possible, discerning water quality trends on a lake-specific basis will often require that a long-term monitoring program be implemented at the local level.

Water quality data is essential to understanding a lake and in making sound management decisions. If your lake is not currently being sampled, it might be time to start. With this being the 50th year of The Michigan Riparian, we are reminded to ask "What will the trophic state of our lakes be 50 years from now?" In the absence of water quality data, we will not know.

Sampling Programs

The 2001-2010 data set previously discussed was collected as part of Michigan's Lake Water Quality Assessment (LWQA) monitoring program in which all lakes greater than 25 acres with developed public boat launches were monitored. The LWQA is a joint monitoring program between the MDEQ and the USGS. With this program, lakes are monitored on a 5-year rotational basis across Michigan's 45 watershed management units. Each year, 7 to 10 of the major watershed units are monitored.

In addition to Michigan's LWQA monitoring program, there are several other sampling initiatives in Michigan. Michigan's Cooperative Lakes Monitoring Program (CLMP) is the second-longest-running volunteer monitoring program in the country. CLMP began in 1974, and data is currently being collected annually from about 250 lakes state-wide. CLMP volunteers monitor Secchi transparency, total phosphorus, chlorophyll-a, dissolved oxygen and temperature, and aquatic plants. These data are used to document baseline water quality conditions and long-term trends in water quality in individual lakes. The CLMP is administered jointly by the Michigan Lake and Stream Associations, Inc. and the Michigan Department of Environmental Quality. Key partners in CLMP include the Great Lakes Commission, the Huron River Watershed Council, and Michigan State University's Department of Fisheries and Wildlife. Customized reports of long-term trends are prepared for individual lakes enrolled in the program.

Another important monitoring effort is the Environmental Protection Agency (EPA) National Lakes Assessment (NLA). The first-ever NLA was conducted in 2007 and included a total of 1,028 lakes across the lower 48 states, including 50 lakes in Michigan (U.S. Environmental Protection Agency 2010). The NLA used a statistical survey design in which lakes were randomly selected to represent the conditions of the larger population of lakes across the lower 48 states. Uniform sampling techniques were used in the assessment so that comparisons could be made between different regions of the country. A separate analysis on Michigan NLA data was prepared (Bednarz 2011). In terms of trophic classification, over 80 percent of Michigan lakes were oligotrophic or mesotrophic (Bednarz 2011). These findings were comparable to the larger 2001-2010 data set collected as part of Michigan's LWQA. Compared to lakes nationally, Michigan had a higher percentage of oligotrophic and mesotrophic lakes and fewer eutrophic lakes. The EPA conducted a second NLA in 2012 that included 904 lakes in the lower 48 states. The results of the second assessment are pending.

In 2002, the Michigan Department of Natural Resources Fisheries Division implemented the Status and Trends Program (STP). This program is designed to help address statewide fisheries management needs and to evaluate the status of aquatic habitat and fisheries communities across the state. As part of the STP, baseline water quality data are being collected from a number of lakes.

What You Can Do

Phosphorus is the nutrient that most often stimulates excessive growth of aquatic plants and algae, leading to a variety of problems collectively known as eutrophication. Of the major nutrient pollutants, phosphorus is most amenable to control through management practices. The cumulative impact of shoreline development is likely the major driving force in how quickly many lakes are aging and becoming more eutrophic. What can you do? Curtail the use of fertilizers (especially fertilizers containing phosphorus), and establish and maintain as much natural shoreline on your property as possible.

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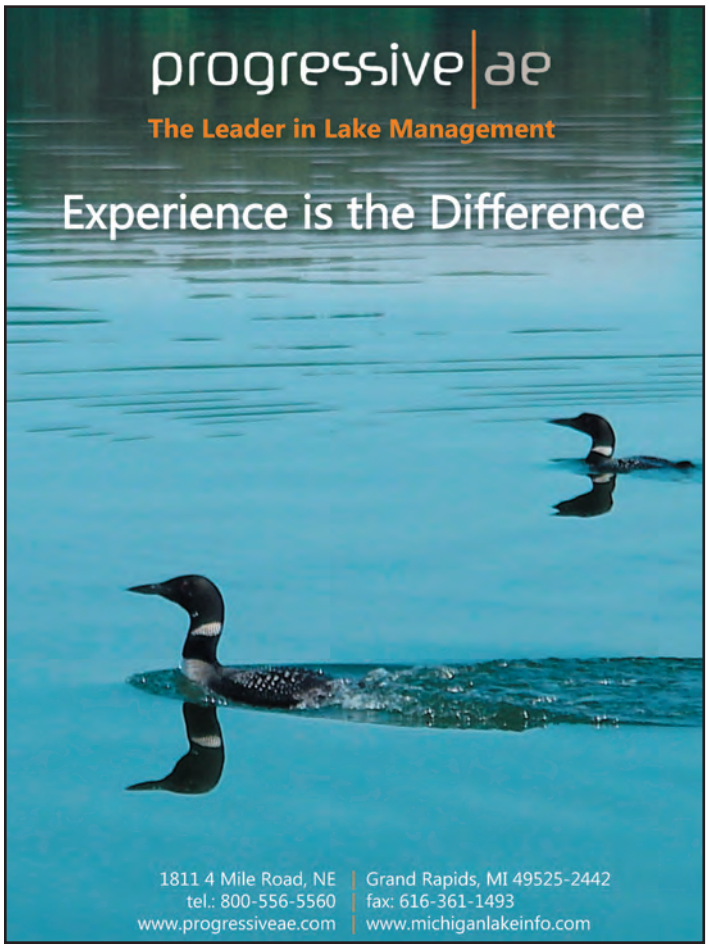

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With phosphorus, an ounce of prevention is worth a pound of cure.

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Swimmer's Itch Research Project

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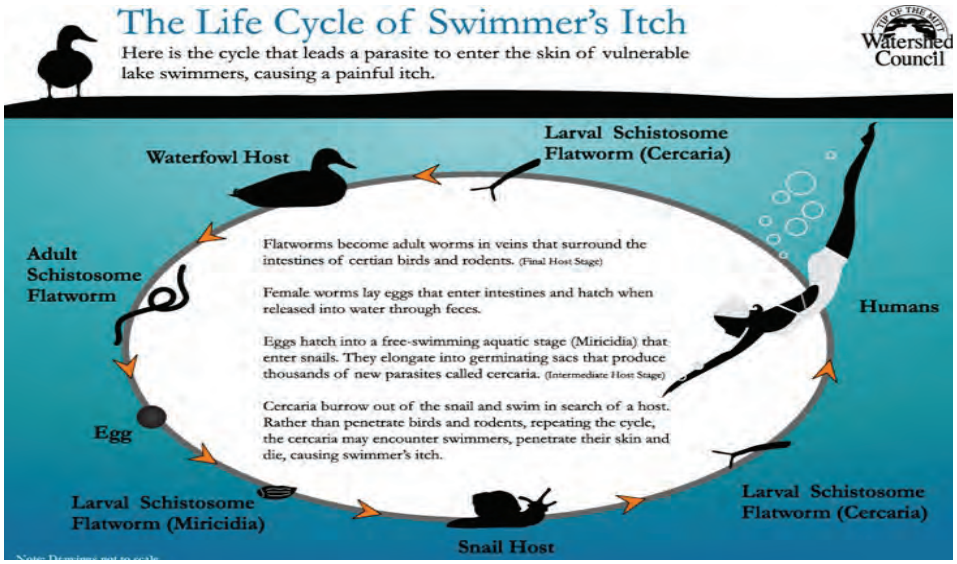

Life Cycle of Swimmers Itch - Watershed Council

Swimmer's itch is a nasty rash caused by parasitic flatworms in the family Schistosomatidae, which normally infect aquatic snails and water birds at various stages of their life cycle. Humans are exposed when infected host snails release tens of thousands of cercariae (larval parasites) which swim through the water to find and penetrate the skin of a definitive host. The cercariae mistakenly burrow into human skin, causing itchy lesions. Many lakes across Michigan are facing the consequences of swimmer's itch which causes intense discomfort, discourages recreational water use, and has economic impacts for regions that rely heavily on lake tourism.

Madelyn Messner is a graduate student in Dr. Thomas Raffel's ecology lab at Oakland University, and will be conducting her thesis work on the environmental variables that drive swimmer's itch exposure in Michigan. Plans for the project involve tracking snail population densities and cercariae abundance eight lakes to better understand and predict the daily fluctuations of cercariae exposure. The lab also hopes to use a DNA detection technique called quantitative PCR to quantify cercariae abundance in water samples. Messner will also measure other variables that may be important predictors of swimmer's itch such as nutrient levels, water temperature, pollutant levels, algae and aquatic vegetation, on-shore wind velocity, bird visitation, and other habitat assessment parameters. After receiving widespread interest in the project from lakes across the state, it may be possible to expand the study to include a comprehensive spatial survey in future years. This type of statewide data could be extremely valuable in understanding and managing the variables that drive swimmer's itch exposure.

Currently the Raffel Lab is supporting this research through limited internal funds and has not secured an external source of funding to cover the costs of reagents, supplies, travel, and personnel for a statewide survey. The lab will be receiving financial support from lake associations to offset the costs of survey work and sample analyses at several lakes in Benzie and Leelanau counties including Crystal Lake, Glen Lake, Lake Leelanau, and Platte Lake; other sites may include Higgins (Roscommon County), Little Traverse, Lime, and Walloon Lakes. Expanding the survey to additional lakes will depend on local support for the project from lake associations, conservation groups, volunteers, and researchers. Outcomes of the project will include statistical analyses and reports, publications in scientific journals, and presentations at relevant conferences, all of which will benefit management and control efforts.

Please contact us if you or your lake have questions about this project or would like to make a donation to offset the costs of swimmer's itch research in the Raffel Ecology Lab.

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